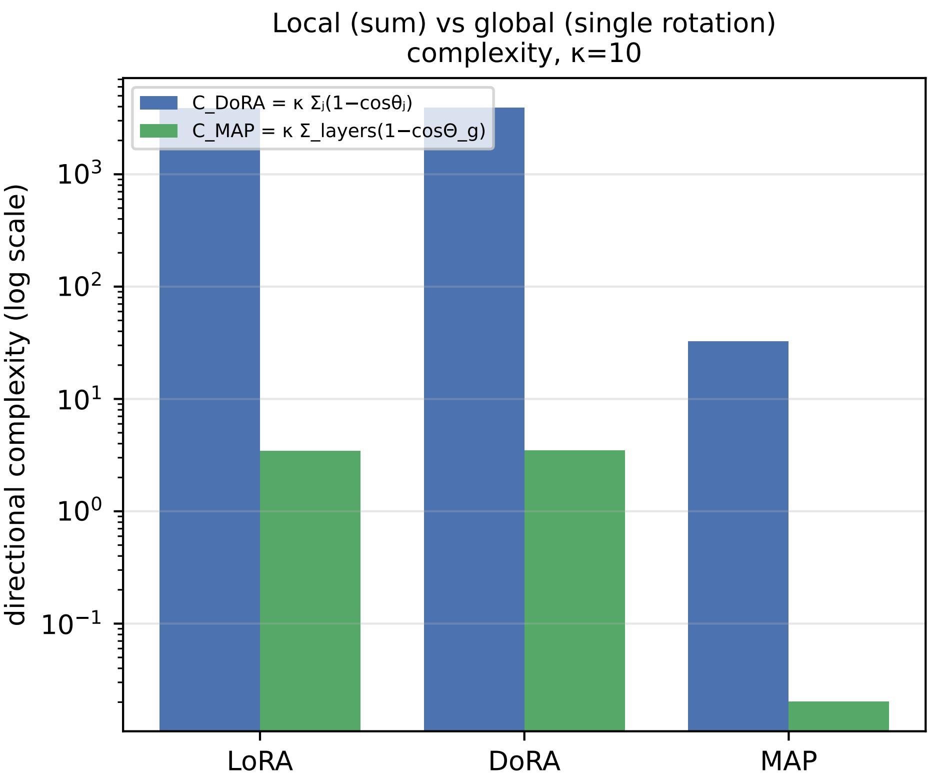
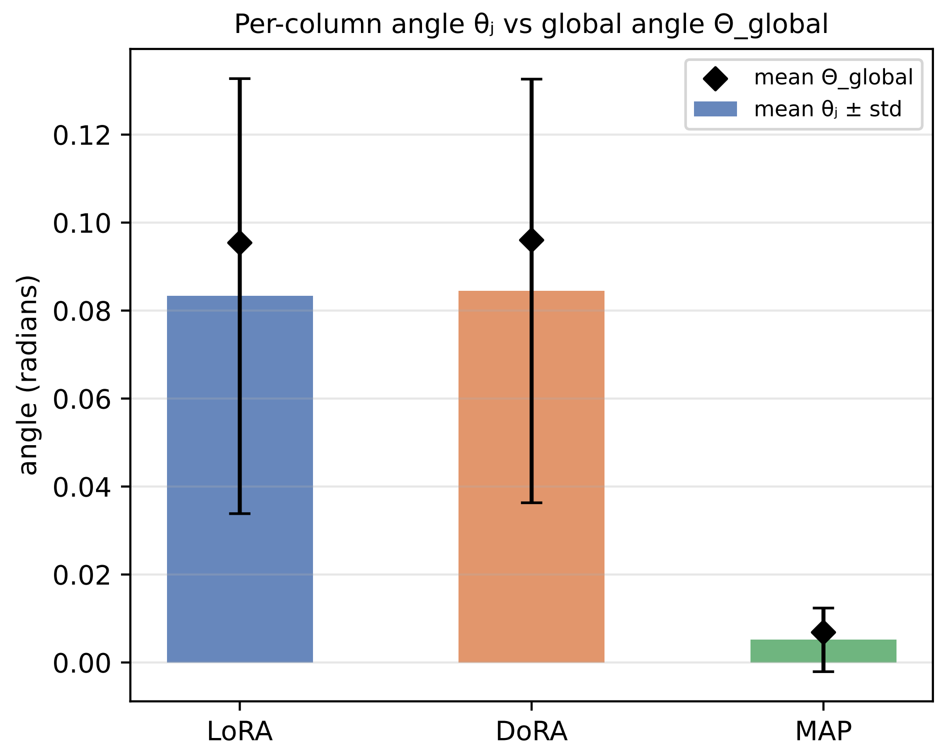
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / MNLI

Fine-tuning metrics and aggregate directional complexity

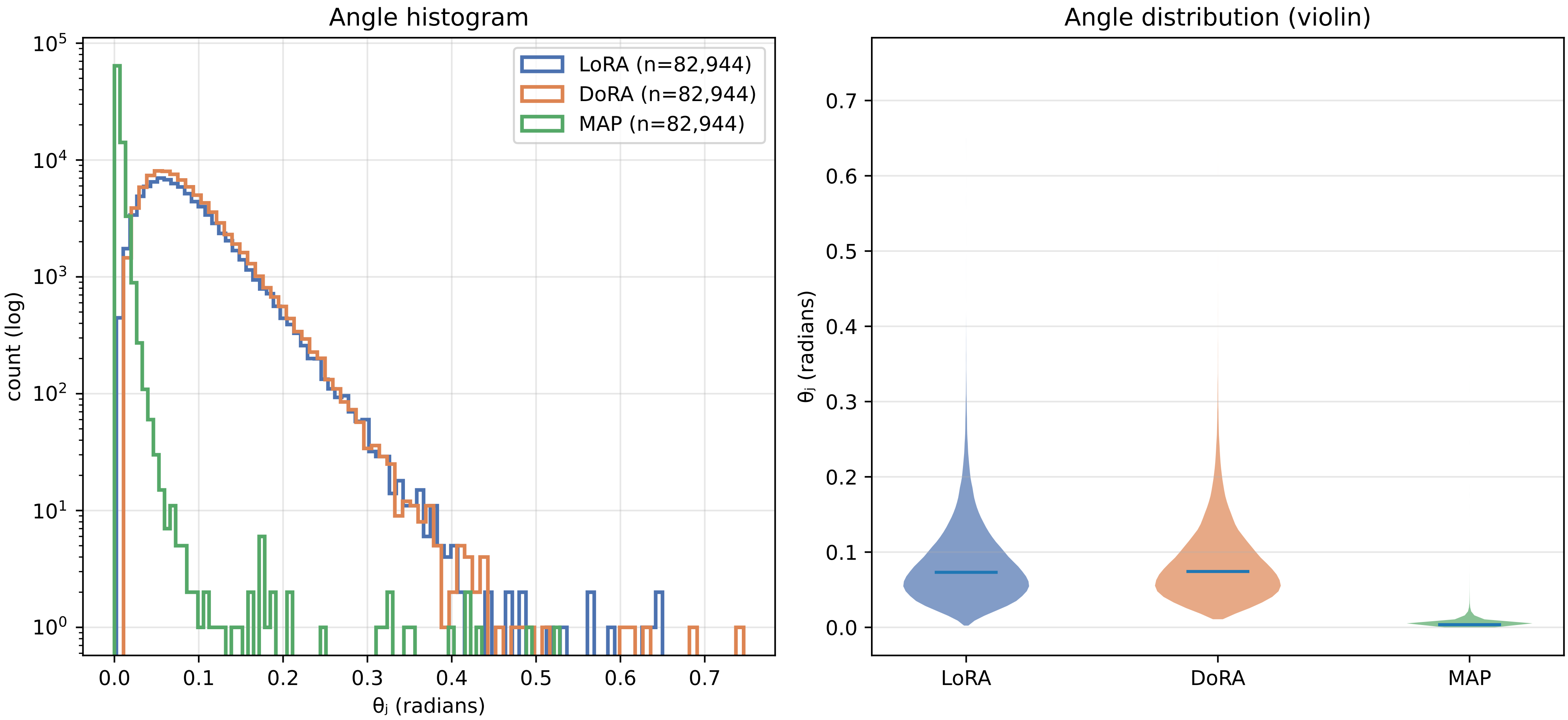
Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	87.71%	1,920,003	3881.91	3.461	0.0833	0.0954	0.997
DoRA	87.74%	2,002,947	3911.60	3.500	0.0845	0.0960	1.000
MAP	85.13%	1,920,147	32.49	0.020	0.0051	0.0068	0.121



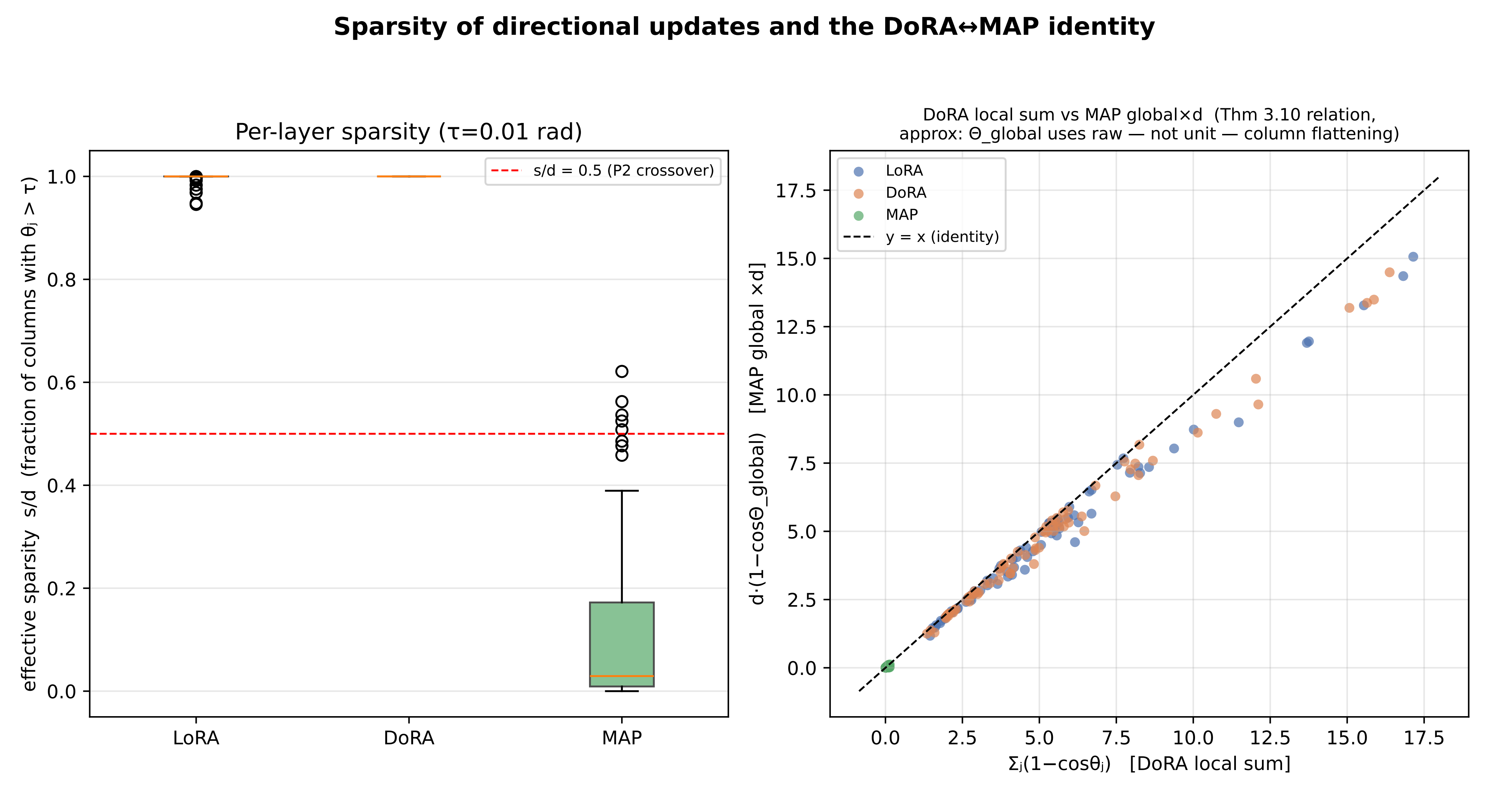


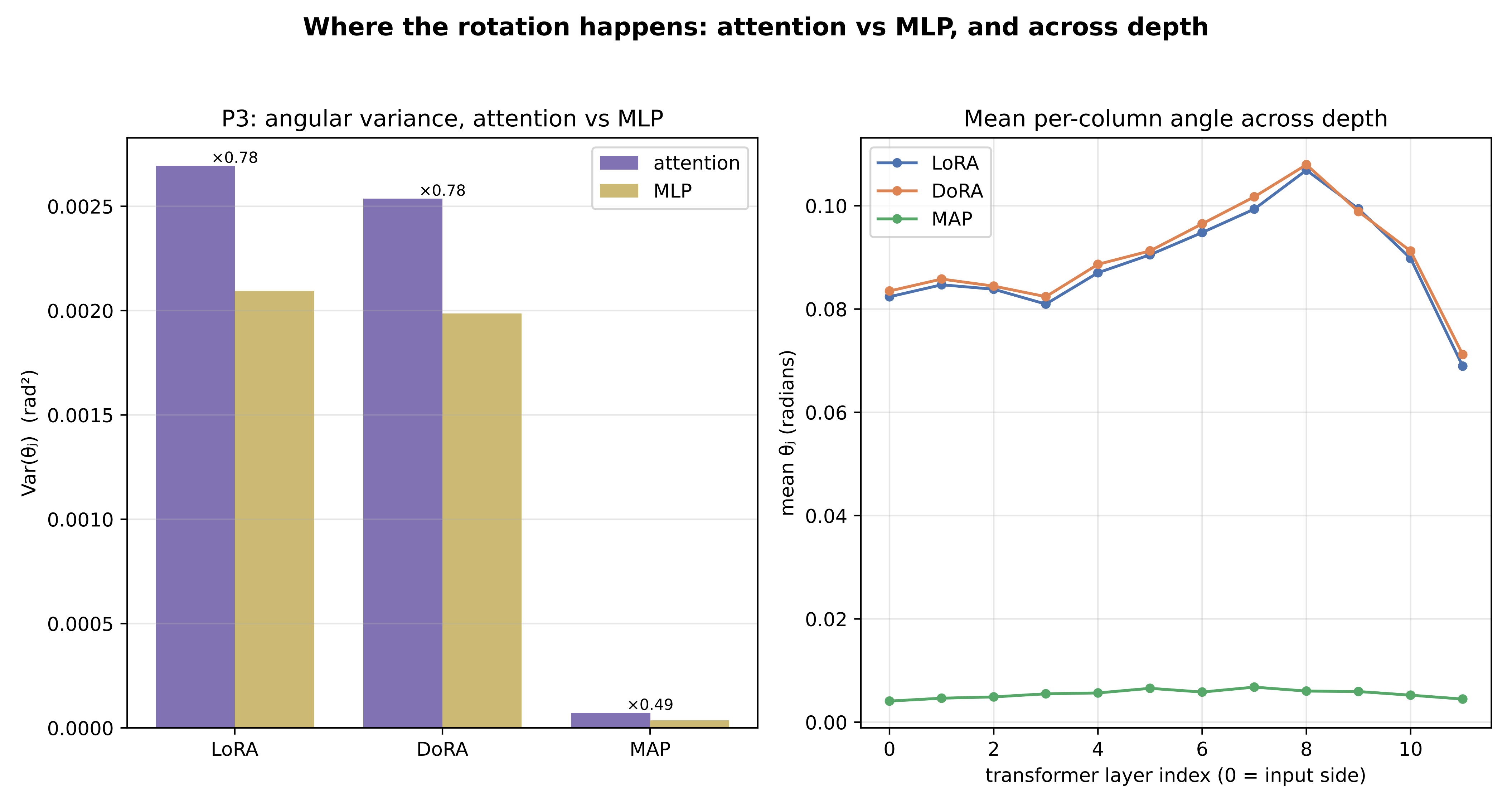
Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta_{global} = \text{mean}_j(1-\cos\theta_j)$ → DoRA sums what MAP averages.

Distribution of per-column directional change θ_j (pooled over all target layers)



A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.


Sparsity of directional updates and the DoRA↔MAP identity
Per-layer sparsity (τ=0.01 rad)
effective sparsity s/d (fraction of columns with θⱼ > τ)
s/d = 0.5 (P2 crossover)
1.0
0.8
0.6
0.4
0.2
0.0
LoRA
DoRA
MAP
DoRA local sum vs MAP global×d (Thm 3.10 relation,
approx: Θ_global uses raw — not unit — column flattening)
LoRA
DoRA
MAP
y = x (identity)
d·(1−cosΘ_global) [MAP global ×d]
Σⱼ(1−cosθⱼ) [DoRA local sum]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Where the rotation happens: attention vs MLP, and across depth
P3: angular variance, attention vs MLP
attention
MLP
×0.78
×0.78
×0.49
Var(θj) (rad²)
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
LoRA
DoRA
MAP
Mean per-column angle across depth
LoRA
DoRA
MAP
mean θj (radians)
0.00
0.02
0.04
0.06
0.08
0.10
0
2
4
6
8
10
transformer layer index (0 = input side)